

Embedded systems industry offers great amount of career opportunities in various fields that are connected with product life-cycle and typical system architectures of current embedded products, informs Halappa. Basic requirement for a career aspirant in this field would be having specialisation in electronics, computers or any other related engineering. He says, "If we have to rank various fields by the number

of opportunities generated by them, the list would be topped by embedded software design (~73%), followed by ASIC design (~15%), FPGA design (~4%), board design (~4%), CAD design (~2%), production services (1%) and sales and marketing (~1%)."

How freshers are placed

To design even a simple product, one has to study and know so many parameters and design requirements, which cannot be achieved without experience and cannot be taught at college, opines Gohel. He says, "Generally fresher engineers are considered to be 'too early' for embedded product design. So, for the initial few months, we put them under testing and validation of designs, and after they gather some experience, we let them develop customer applications under the supervision of senior engineers." "The semiconductor industry always comes up with new and improved components, so one has to keep updating with new technologies used and market's needs. This industry is only for the passionate people, and hence one cannot survive if s/he is not abreast with the latest happenings," he adds.

On the positive side, there is a good scope for fresh engineering talent in the industry, provided they meet the basic industry standards, informs VEDI. VVDN's selection process consists of basic-level (C/C++, electronics, aptitude) written entrance exam followed by an interview. She says, "Entry-level role is normally that of a trainee. They

Expectations from freshers

Technical skills

1. Basic electronics, logic design and microcontroller
2. Microprocessor and controller architecture, and Assembly language
3. Analogue circuits, linear electronics and digital electronics
4. Programming languages like C, C++, Java and HTML
5. Fundamentals of operation systems, real-time operating systems, inter-process communication, data structures
6. Computer networks

Personal skills

1. Basic aptitude
2. Good communication skills
3. Good analytical and problem-solving skills
4. Critical thinking towards field problems
5. Desire and passion to learn
6. Adaptation to new trends and technologies

are on probation for three months. Students need not go to specialised training centres after their graduation." She adds, "As a trainee, candidates get an opportunity to get involved in various roles, which include hardware development, software development and testing. Based upon the candidates' interest and performance during the probation, trainees are accredited as full-time employees and are given specialised designations based on their interest and company's requirement."

Talking about elfin chips, Kapat informs, "In product design projects, we prefer candidates to start as application engineers and then gradually, when they understand the product, they can dig into lower layers. Candidates who have good knowledge of programming languages and operating systems can directly start as firmware engineer where they get opportunity to learn about different systems on chips (SoCs)." She adds, "We also have well-defined on-the-job training programmes where a fresh candidate not only learns project-specific things but is assigned responsibilities to perform tasks."

Typical training period

Patel shares that it is very difficult to get experienced personnel, and the only way is to hire fresh engineering graduates and train them for the required technologies. He says, "As colleges are lacking qualified lecturers and infrastructure to cope up with new technology development, fresh engineers should start with a few

months of training on basics and devices/technologies used in current products. Once they spend about three to four months, they can be used as design engineers or firmware engineers under the supervision and guidance of senior engineers." He adds, "Even at entry level, they can prepare schematics, small board layouts and write application software modules."

Typically, entry-level engineers would be trainee engineers for three to six months, during which they get exposed to the company's work culture, product design life-cycle, design flow and methodology, standards/guidelines for design, coding, testing and safety, and quality compliance and IP protection policies. Halappa says, "After their tenure as trainee, based on individuals' performance in their earlier project tasks and their background experience, they would be assigned module-level design and/or testing tasks. But the kind of work and exposure that an individual gets may well depend on the dynamics of ongoing projects and also individual's ability to grow with the challenges." He adds, "Typically, fresh engineers would mature as full-fledged design engineers in a span of three to five years, depending on their focus and ability."

Higher demand for software engineers

"With the rise of the Internet of Things, embedded systems have received a huge push forward and their complexity has led to a great demand for skilled professionals and technologists in this field. The majority of the roles are focussed in the IT software sector with Bengaluru, Delhi and Mumbai serving as the main hubs for companies engaged in employing these high-end programming skills. As with all niche specialisations, entry-level salaries quickly rise as the professional gains valuable on-the-job experience and deals with real-world scenarios," explains Vivek Madhukar, COO, Times Business Solutions Limited.